

Amazon RDS (Relational Database Service).

What is Data?

Data is a collection of information. It can be structured, semi-structured, or unstructured.

What is a Database?

A database is a collection of organized data that can be easily accessed, managed, and updated. Databases typically store data in structured formats with rows and columns, following predefined schemas.

Types of Data Storage:

1. Structured Data:

- Data organized into rows and columns with predefined schemas, such as SQL databases (e.g., MySQL, PostgreSQL, Oracle).

2. Semi-Structured Data:

- Data that does not fit into a rigid schema but contains tags or markers to separate data elements, such as JSON or XML formats.

3. Unstructured Data:

- Data that does not have a predefined format or structure, such as videos, images, or text documents.

This type of data is often stored in services like Amazon S3.

Before Cloud Computing:

- Traditionally, businesses had to purchase and manage their own physical servers, operating systems, and software. This often required significant resources and expertise.

With Cloud Computing:

- Cloud services, like AWS, shift the responsibility of hardware and software management to the cloud provider. AWS manages the infrastructure, and you only need to handle the configuration and management of your applications and databases.

AWS RDS (Relational Database Service):

- AWS RDS is a managed database service that simplifies database setup, operation, and scaling. AWS handles the underlying hardware, database software, and administrative tasks such as backups, patching, and scaling.
- It supports several database engines, including MySQL, PostgreSQL, Oracle, SQL Server, and Amazon Aurora.

Database Management with AWS RDS:

1. Creating a Database:

- Use the AWS Management Console to create a new RDS instance based on your configuration needs.
- During the creation process, you will be provided with credentials (username and password). Make sure to save these credentials securely as they are shown only once.

2. Connecting to RDS:

- You can connect to your RDS instance using various database management tools such as MySQL Workbench, SQL Server Management Studio (SSMS), Sqlectron or DBeaver. Download and install your preferred client tool.
- To connect, enter the database name, type, server address, username, and password into the tool.

3. Managing Backups and Snapshots:

- AWS RDS automatically creates backups and snapshots of your database. It's important to review and manage these backups to ensure data recovery and compliance with your backup policies.

4. Practicing Queries:

- Invest time in practicing basic queries and operations. There are many resources available, including tutorials on platforms like YouTube, to help you get familiar with querying your database.

Other Cloud Providers:

- Similar managed database services are offered by other cloud providers, each with its own set of features and service names.